

An Equal-Action Node and a Hill–Kummer Class in the Period-Five Area-Preserving Hénon Family

Anonymous Research Note

12 August 2026

Abstract

We prove that the exact period-five action map of the area-preserving Hénon family contains an arithmetic stability invariant that is absent from its known orbit permutation cover. Starting from the chronological recurrence, we derive the six-sheet period-five normalization and its cyclic action image. The action discriminant has a coprime degree-nine factor P_9 . Over $K_9 = \mathbb{Q}[A]/(P_9)$, the repeated action value has exactly two distinct normalization points; their image is a transverse ordinary node. Both points have exact period five, their action Hessians are nondegenerate, and their return matrices have neither multiplier $+1$ nor -1 . If h_1, h_2 denote their chronological Hill determinants, then the product $N_H = h_1 h_2$ descends through branch exchange and defines a normalization-invariant square class. Its exact rational field norm is

$$N_{K_9/\mathbb{Q}}(N_H) = \frac{2^6 13 19^5 41 59^5 5653^2}{3^5},$$

so N_H is not a square in K_9 . Hence $u^2 = N_H$ is a nontrivial quadratic Kummer extension of the generic collision field. We also prove $\text{Gal}(P_9/\mathbb{Q}) = S_9$ and release an independently checked exact certificate. The result is fixed-period arithmetic structure, not a dynamical zeta function or a Hilbert–Pólya operator.

1 Introduction

An orbit equation records where periodic trajectories live, but it need not record how their linearized dynamics differ. This loss is acute when two periodic points have the same value of a generating action: the action image identifies the two points, while their return matrices may remain different. The present paper shows that this apparent information loss produces a canonical arithmetic decoration in a concrete area-preserving Hénon family.

We study

$$H_A(q, p) = (1 - Aq^2 - p, q), \tag{1}$$

in the area-preserving quadratic tradition initiated by Hénon [6], with a diagonal reversor-line initial state. Its period-five critical points are encoded by a cyclic action $\Phi_{5,A}$. Eliminating the orbit coordinate gives a plane curve $W_5(A, c) = 0$, where c is the action value. The normalization of this curve is not new: an exact linear subresultant recovers the familiar six-sheet period-five marker cover. The new information appears where the action map fails to separate two points.

The action discriminant contains a degree-nine factor $P_9(A)^2$ that is coprime to the ramification factors of the normalization. A squared discriminant factor alone does not determine the singularity. We therefore work over the generic collision field and reconstruct the normalization fiber, tangent cone, and two normalization slopes. The fiber is a separable quadratic and the slopes differ, proving that the action image has an ordinary node.

The stability data then descend in a form that the bare action coordinate does not display. Let

$$h_i = \det(I - DH_A^5(z_i))$$

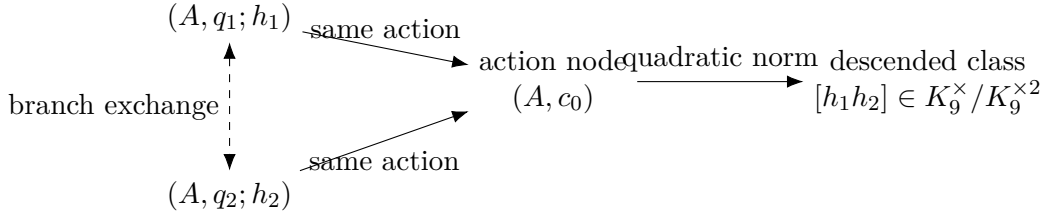


Figure 1: Two distinct points of the period-five normalization map to one ordinary action node. Their individual Hill values are branch-labelled, but their product descends and defines a nontrivial square class.

at the two points above the node. Branch exchange fixes h_1h_2 , and a common normalization of both Hill values changes the product by a square. The resulting class in $K_9^\times/K_9^{\times 2}$ is nontrivial, as shown by its exact rational field norm. Figure 1 summarizes the construction.

The contributions are precise and falsifiable:

1. We derive the action curve from the chronological recurrence and separate the known normalization from its singular action embedding.
2. We prove that the generic degree-nine collision is a two-branch transverse ordinary node formed by distinct exact-period-five points.
3. We prove the action-Hessian/Hill identity, exclude both $+1$ and -1 multipliers, and descend the symmetric Hill square class.
4. We prove that this class is nonsquare by an exact rational norm and prove $\text{Gal}(P_9/\mathbb{Q}) = S_9$ from three modular cycle types.

Each generic ingredient has prior art, and we make that boundary explicit. The result supplies no all-period clock, trace formula, analytic determinant, or self-adjoint operator. Its value is narrower: it is a theorem-level example in which chronological stability adds a nontrivial arithmetic layer to a singular Hénon action map.

2 Prior work and claim boundary

Exact Hénon orbit algebra. Exact low-period equations, orbital sums, discriminants, reversibility classes, and Galois groups for the area-preserving Hénon map have been developed in detail by Endler and Gallas [4, 5]. Polynomial bridges can make apparently different orbit equations birationally or algebraically equivalent [3]. These results prevent us from treating the period-five sextic or its ordinary permutation group as a new cover. Our linear inverse subresultant makes the equivalence explicit; the contribution lies in the action-image singularity and the stability data above it.

Generating actions and equal-action saddles. Periodic action principles for reversible symplectic maps are classical [7]. In the quantum-Hénon setting, equal-action saddles already enter Stokes geometry and virtual turning-point constructions [9]. We therefore do not claim the first Hénon action or the first equal-action Hénon pair. Our setting differs in being a closed exact-period-five family over the parameter line, with an exact collision polynomial and a return-stability square class.

Critical-value geometry. Discriminants of critical values and their monodromy belong to the Lyashko-Looijenga framework [8]. In that convention, the locus where distinct Morse points

Object	Status before this work	Exact increment here
Period-five marker cover	known	birationality rederived as a control
Discrete Hénon action	known mechanism	exact action plane model W_5
Equal-critical-value locus	generic prior art	explicit coprime factor P_9
Plane node	generic prior art	two-point/tangent proof on P_9
Hill formula	known theorem	chronological Hénon specialization
Kummer cover	known mechanism	nonsquare class $[h_1 h_2]$

Table 1: The novelty boundary. The contribution is a coupled exact specialization, not any generic mechanism in isolation.

share a critical value is a Maxwell stratum; some real catastrophe-theory conventions reserve “Maxwell set” for equal global minima. We use the neutral phrase *equal-action node* and make no minima claim. The generic occurrence of a node or a symmetric critical-value group is not new. What is specific here is the explicit Hénon polynomial and its chronological Hill decoration.

Hill and Kummer structures. The equality between a discrete action Hessian and a monodromy determinant is a specialization of Hill’s formula [2]. Kummer covers and their braid monodromy are likewise established [1]. We use these as structural boundaries: neither mechanism is claimed as new, and an ordinary node does not automatically supply a Picard–Lefschetz representation. The theorem proved below is the coupled specialization summarized in Table 1.

3 Chronological action curve

Set $(x_0, x_{-1}) = (q, q)$ and iterate

$$x_{i+1} = 1 - Ax_i^2 - x_{i-1}. \quad (2)$$

The state after five steps is (x_5, x_4) . Taking the gcd of $x_5 - q$ and $x_4 - q$ over $\mathbb{Q}(A)[q]$, removing the fixed-point factor $Aq^2 + 2q - 1$, and primitive-scaling yields the sextic $G_A(q)$ printed in Appendix A. Its discriminant is

$$\text{Disc}_q G_A = 2^6 A^{30} P_2 P_5, \quad (3)$$

where

$$P_2 = 16A^2 - 28A - 19, \quad P_5 = 16A^5 - 108A^4 + 105A^3 + 27A^2 - 97A - 47.$$

Because five is prime, removing the fixed-point factor leaves exact period five away from $AP_2 P_5 = 0$.

The cyclic action is

$$\Phi_{5,A} = \sum_{i \bmod 5} \left(x_i x_{i+1} - x_i + \frac{A}{3} x_i^3 \right). \quad (4)$$

Its Euler–Lagrange equations are

$$\partial_{x_i} \Phi_{5,A} = x_{i-1} + x_{i+1} - 1 + Ax_i^2 = 0,$$

which reproduce (2). Reducing $3A^2 \Phi_{5,A}$ modulo G_A gives a degree-five polynomial $R(A, q)$. We define the primitive action image by

$$W_5(A, c) = A^{-30} \text{Res}_q(G_A(q), 3A^2 c - R(A, q)). \quad (5)$$

Proposition 3.1 (Birational control). *The plane curve $W_5(A, c) = 0$ and the normalization $G_A(q) = 0$ have the same function field over $\mathbb{Q}$.*

Proof sketch. At $A = 6$, reduction modulo 37 preserves degree six. The coefficient rows of G_A and W_5 , from highest to lowest degree, are

$$(36, 12, 21, 21, 26, 27, 34), \quad (26, 34, 7, 16, 19, 8, 4),$$

and exact finite-field factorization gives profile (6) for each. Consequently both generic sextics are irreducible over $\mathbb{Q}(A)$. The last nonzero subresultant of $G_A(q)$ and $3A^2c - R(A, q)$, viewed as polynomials in q , is linear:

$$U(A, c)q + V(A, c).$$

Its coefficient U is coprime to W_5 over $\mathbb{Q}(A)$. Therefore $q = -V/U$ on a dense open set, proving $\mathbb{Q}(A, q) = \mathbb{Q}(A, c)$. The released certificate stores and independently checks the complete coefficient ledgers. $\square$

This proposition is a negative control with a positive consequence. It rules out a new ordinary cover, but it also identifies singularities of W_5 as failures of the action coordinate to separate points on the known normalization.

4 The degree-nine equal-action node

Exact elimination gives

$$\text{Disc}_c W_5 = 2^{12} 3^{30} A^{60} P_2^5 P_5^3 P_9^2, \quad (6)$$

where

$$\begin{aligned} P_9(A) = & 110592A^9 - 294912A^8 + 159744A^7 + 225792A^6 \\ & - 162816A^5 - 51520A^4 + 50672A^3 + 736A^2 - 6032A + 1037. \end{aligned} \quad (7)$$

The polynomial P_9 is squarefree and coprime to AP_2P_5 . Hence it does not arise from the normalization ramification in (3).

Let $K_9 = \mathbb{Q}[A]/(P_9)$. In $K_9[c]$, the gcd of W_5 and $\partial_c W_5$ is $c - c_0$. Substituting c_0 into the normalization equations produces

$$g_2(q) = \gcd(G_A(q), 3A^2c_0 - R(A, q)) = q^2 + bq + d. \quad (8)$$

The discriminant $b^2 - 4d$ is nonzero, and g_2 is coprime to G_A/g_2 . Thus the repeated action value has exactly two distinct normalization points.

Theorem 4.1 (Ordinary node). *At the generic point of $P_9 = 0$, the action image $W_5 = 0$ has a transverse ordinary node whose two branches are the two points cut out by (8).*

Proof sketch. Exact evaluation in K_9 gives

$$W_5 = (W_5)_A = (W_5)_c = 0, \quad (W_5)_{cc} \neq 0,$$

and

$$(W_5)_{Ac}^2 - (W_5)_{AA}(W_5)_{cc} \neq 0. \quad (9)$$

Therefore the first nonzero homogeneous term is a nondegenerate quadratic. To identify its two tangents with the two normalization branches, differentiate $G(A, q) = 0$ and $c = R/(3A^2)$. The branch slope is

$$s(q) = \frac{(A\partial_A R - 2R)\partial_q G - A(\partial_q R)\partial_A G}{3A^3\partial_q G}. \quad (10)$$

The denominator is invertible modulo g_2 . Reduction gives $s(q) \equiv uq + v$, with

$$(s(q_1) - s(q_2))^2 = u^2(b^2 - 4d) \neq 0.$$

The two normalization branches therefore have distinct image tangents. $\square$

The collision parameters themselves have a large arithmetic permutation group.

Proposition 4.2. $\text{Gal}(P_9/\mathbb{Q}) \cong S_9$.

Proof. The exact unramified factor degrees modulo 7, 17, 23 are respectively (9), (5, 2, 1, 1), and (8, 1). The first makes P_9 irreducible. The 8-cycle makes a point stabilizer transitive on the other eight letters, so the group is 2-transitive and primitive. Squaring an element of type (5, 2, 1, 1) gives a pure 5-cycle. Jordan's theorem gives A_9 , while the 8-cycle is odd. $\square$

5 Hill–Kummer descent

At a chronological orbit point x_i , write

$$D_i = \begin{pmatrix} -2Ax_i & -1 \\ 1 & 0 \end{pmatrix}, \quad M = D_4 D_3 D_2 D_1 D_0. \quad (11)$$

Every D_i has determinant one. Define the Hill value

$$h_A(q) = \det(I - M) = 2 - \text{tr } M.$$

The cyclic Hessian of (4) has diagonal entries $2Ax_i$ and cyclic nearest-neighbor entries one. Direct exact expansion gives the Hénon specialization of Hill's formula,

$$\det D^2 \Phi_{5,A} = h_A(q) \quad \text{on } G_A(q) = 0. \quad (12)$$

The resultant

$$\text{Res}_q(G_A, h_A) = -2^6 A^{30} P_2^5 P_5 \quad (13)$$

is coprime to P_9 . Thus both node branches are Morse and exclude multiplier $+1$.

Morse nondegeneracy does not exclude multiplier -1 . Because $\det M = 1$,

$$\det(I + M) = 2 + \text{tr } M = 4 - h_A.$$

The separate exact gate is

$$\text{Res}_q(G_A, 4 - h_A) = -2^6 A^{30} B_6 B_9, \quad (14)$$

where B_6, B_9 are printed in Appendix A and both are coprime to P_9 . Consequently neither branch has multiplier $+1$ or -1 .

Now reduce $h_A(q)$ in the quadratic algebra $E = K_9[q]/(g_2)$:

$$h_A(q) \equiv \ell q + m.$$

Let $\bar{q}$ be the class of q in E and let σ denote the exchange involution. Using $q_1 + q_2 = -b$ and $q_1 q_2 = d$, the quadratic norm of $h = h_A(\bar{q})$ is

$$N_H = N_{E/K_9}(h_A) = h_1 h_2 = \ell^2 d - \ell m b + m^2. \quad (15)$$

Lemma 5.1 (Descent and gauge). *The square class $[N_H] \in K_9^\times / K_9^{\times 2}$ is fixed by branch exchange, by cyclic action coboundaries, by parameter-dependent action translations or nonzero rescalings, and by a common Hill normalization. Moreover, $[h/\sigma(h)] = [h\sigma(h)]$ in $E^\times / E^{\times 2}$; over a splitting field this reads $[h_1/h_2] = [h_1 h_2]$.*

Proof. Branch exchange fixes the product. A cyclic coboundary telescopes on a closed orbit, while action translation or rescaling changes only the target coordinate and not the return matrices. A common change $h_i \mapsto \nu h_i$, with $\nu \in K_9^\times$, multiplies $h_1 h_2$ by ν^2 . Finally,

$$\frac{h/\sigma(h)}{h\sigma(h)} = \sigma(h)^{-2}.$$

Independent branch-dependent rescalings are not common dynamical normalizations and are excluded from the gauge group. $\square$

Theorem 5.2 (Nontrivial Hill–Kummer class). *The element N_H is not a square in K_9 . Hence $u^2 = N_H$ defines a nontrivial quadratic extension of the generic collision field.*

Proof. Exact quotient arithmetic gives

$$N_H = -\frac{4}{4827099043} Q_8(A) \quad \text{in } K_9,$$

where Q_8 is printed in Appendix A. The leading coefficient of P_9 is 110592, so the nonmonic norm formula is

$$N_{K_9/\mathbb{Q}}(N_H) = \left(-\frac{4}{4827099043}\right)^9 \frac{\text{Res}(P_9, Q_8)}{110592^8}.$$

It simplifies to

$$N_{K_9/\mathbb{Q}}(N_H) = \frac{2^6 13 19^5 41 59^5 5653^2}{3^5}. \quad (16)$$

The rational number is not a square. A square in K_9 would have square rational norm, giving a contradiction. $\square$

Theorem 5.2 proves one nontrivial quadratic class. It does not prove independence of its nine parameter conjugates or a full $C_2 \wr S_9$ Galois group.

6 Finite controls and reproducibility

The characteristic-zero norm (16) is the primary nonsquare proof. Finite reductions test the specialization and catch branch, chronology, and square-class errors. We froze every prime dividing

$$P_9(6) = 61 \cdot 157 \cdot 3203 \cdot 21943$$

before the Phase-3 theorem computation. Table 2 reports the complete control set.

p	action	q_1, q_2	h_1, h_2	$\chi_p(h_1 h_2)$	role
61	45	12, 36	44, 7	+1	post-hoc C32 regression
157	76	58, 129	26, 96	+1	square control
3203	150	1805, 2375	1577, 2906	−1	nonsquare control
21943	20277	7923, 20203	4539, 16878	+1	square control

Table 2: All prime divisors of $P_9(6)$. The first row was discovered in C32 and remains explicitly post-hoc; no finite row proves Theorem 5.2.

The released producer derives all promoted polynomials from (1)–(4). It serializes rational polynomials and elements of $\mathbb{Q}[A]/(P_9)$ canonically. The checker imports no producer module. It reconstructs the recurrence, action, resultants, collision fiber, tangents, derivative chronology, Hessian, field norm, and modular controls, then enforces exact key sets and type-strict JSON equality.

The independent report passes 12 semantic gates. Thirty-three adversarial and regression tests alter the source hash, polynomial data, discriminant exponent, irreducibility or Galois ledgers, collision fiber, tangent, -1 -multiplier gate, field norm, finite controls, post-hoc disclosure, Route-A status, and scope. Expected mathematical mismatches are distinguished from unexpected checker errors. The full project is frozen by a SHA-256 manifest and reproduced by a read-only default runner.

These checks do not make two computer-algebra implementations logically independent proof assistants. The explicit derivation and mutation suite instead provide a transparent, reproducible exact certificate whose main identities can be checked outside the producer.

7 Discussion and Route-A boundary

The action coordinate identifies two period-five points, but chronological linear stability does not disappear at the identification. Its symmetric product descends and remains nonsquare. This is the paper’s structural conclusion: a singular action embedding can carry an arithmetic stability cover that is invisible to the ordinary orbit permutation cover.

The result also narrows the next mathematical question. Proposition 4.2 provides nine conjugate collision parameters, while Theorem 5.2 proves only one nontrivial square class over their generic field. Determining the rank of the conjugate classes would decide whether the quadratic extension grows into a larger Kummer module or collapses through hidden relations. A different continuation would seek compatible classes at varying periods and one trace law that preserves the chronological action and Hill data.

Neither continuation has been proved here. In particular, a finite algebraic cover does not define a primitive-orbit Euler product, and its Galois group does not imply analytic continuation, a functional equation, or a critical-line spectrum. Under the project’s strict evaluation, the current ceiling is

$$(A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT),$$

with overall decision `ROUTE_A_REJECTED`. Route B is not authorized because no self-adjoint operator has been constructed.

The fixed-period limitation is therefore both a boundary and a guide. The next large advance must prove an all-conjugate or all-period theorem; another isolated prime or longer low-period ledger would not close the missing logical bridge.

References

- [1] Enrique Artal Bartolo, José Ignacio Cogolludo-Agustín, and Jorge Ortigas-Galindo. Kummer covers and braid monodromy. *Journal of the Institute of Mathematics of Jussieu*, 13(3): 633–670, 2014. doi: 10.1017/S1474748013000297.
- [2] Sergey Bolotin and Dmitry Treschev. Hill’s formula. *Russian Mathematical Surveys*, 65(2): 191–257, 2010. doi: 10.1070/RM2010v065n02ABEH004671.
- [3] Owen J. Brison and Jason A. C. Gallas. Polynomial interpolation as detector of orbital equation equivalence. *International Journal of Modern Physics C*, 29(8):1850096, 2018. doi: 10.1142/S0129183118500961.
- [4] Antônio Endler and Jason A. C. Gallas. Arithmetical signatures of the dynamics of the Hénon map. *Physical Review E*, 65(3):036231, 2002. doi: 10.1103/PhysRevE.65.036231.
- [5] Antônio Endler and Jason A. C. Gallas. Reductions and simplifications of orbital sums in a hamiltonian repeller. *Physics Letters A*, 352(1–2):124–128, 2006. doi: 10.1016/j.physleta.2006.01.031.

- [6] Michel Hénon. Numerical study of quadratic area-preserving mappings. *Quarterly of Applied Mathematics*, 27(3):291–312, 1969. doi: 10.1090/qam/253513.
- [7] H.-T. Kook and James D. Meiss. Periodic orbits for reversible, symplectic mappings. *Physica D: Nonlinear Phenomena*, 35(1–2):65–86, 1989. doi: 10.1016/0167-2789(89)90096-1.
- [8] Eduard Looijenga. The complement of the bifurcation variety of a simple singularity. *Inventiones Mathematicae*, 23(2):105–116, 1974. doi: 10.1007/BF01405164.
- [9] Akira Shudo and Kensuke S. Ikeda. Stokes geometry for the quantum Hénon map. *Nonlinearity*, 21(8):1831–1880, 2008. doi: 10.1088/0951-7715/21/8/007.

A Exact polynomial ledgers

The primitive exact-period-five marker is

$$G_A(q) = A^6 q^6 + 2A^5 q^5 + (-3A^5 + 2A^4)q^4 + (-4A^4 + 2A^3)q^3 \\ + (3A^4 - 4A^3 + A^2)q^2 + (2A^3 - 2A^2)q - A^3 + 2A^2 - A - 1.$$

The reduced action numerator is

$$R(A, q) = 2A^5 q^5 + 2A^5 q^4 + 2A^4 q^4 - 4A^4 q^2 - 2A^3 q^3 + 6A^3 q^2 \\ - 2A^3 q + 2A^3 - A^2 q^2 + 4A^2 q - 8A^2 + 2A + 2.$$

The multiplier-minus-one factors in (14) are

$$B_6 = 4096A^6 - 13312A^5 + 512A^4 + 17344A^3 - 7520A^2 - 16276A - 3251, \\ B_9 = 4096A^9 - 50176A^8 + 206336A^7 - 299584A^6 - 75552A^5 + 396684A^4 \\ + 126657A^3 - 257337A^2 - 204849A - 48843.$$

Their resultants with P_9 are nonzero integers; the exact values are stored in the machine certificate.

The polynomial occurring in the Hill norm is

$$Q_8(A) = 1099287308943360A^8 - 2052594837734400A^7 - 280866909871104A^6 \\ + 2471392536562048A^5 + 359663285550912A^4 - 726673931525216A^3 \\ - 99889575560184A^2 + 74824104729347A + 6180584501363.$$

The complete action polynomial W_5 , collision value c_0 , quadratic fiber g_2 , derivative evaluations, branch-slope coefficients, and resultants are large. They are serialized in canonical coefficient order in `results/c33_kummer_certificate.json`; the independent checker decodes and reconstructs every one of them before accepting the theorem gates.

B Scope of the computational proof

The certificate works over $\mathbb{Q}$, $\mathbb{Q}(A)$, and $\mathbb{Q}[A]/(P_9)$ using exact polynomial arithmetic. Its finite-field rows are derived afterward and serve only as controls. The machine proof does not query a prime table, a Riemann-zero table, or a numerical root finder.

The source-lock includes the staged research question, the prior-work audit, the earlier C32 collision certificate, the chronological Hénon producer used as a convention control, and the foundational area-preserving Hénon manuscript. No coefficient from the Phase-1 pilot is imported as a theorem input.